



**School of Computer Science and Engineering [SCOPE]**

**CAT – I**

**CSE2004 – Database Management System**

**Course Name: B. Tech CSE**

**Slot: B2+TB2**

**Max. Marks: 30**

**SET-1**

**Answer ALL Questions (3\* 10 = 30)**

| Q. No | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max Marks | CO | BL |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|
| 1     | Assume that you are viewing the Database of Goibibo. Explain the role of different types of database users with their activity in the given database.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10        | 1  | L2 |
| 2     | Consider the fruits, vegetables and grocery management system of JioMart. It maintains complete, up-to-date information on sales of various fruits, vegetables and grocery items. The fruits can be portrayed by ID, name, price, and origin. The vegetables can be represented by vegetable ID, name, type, price and description. The grocery can be represented by product ID, name, type, price and description. The sales information can be represented by sales ID, amount, type and quantity. The salesman information can be portrayed by salesman ID, name, address, and contact no. The customer information can be represented by customer ID, name, address, contact no, email ID, purchased amount and payment type. At the end of the day, it is required to calculate how many products have been sold by an individual salesman and their total sales amount. Construct an Entity Relationship diagram to illustrate the above scenario. | 10        | 2  | L3 |
| 3     | Given a relation scheme R(ABCDEFGH) with a set of FDs $F = \{ AB \rightarrow CD, AD \rightarrow G, E \rightarrow D, BC \rightarrow AH, AG \rightarrow EH \}$ .<br><br>(i) Find the minimal cover of the following set of functional dependencies.<br>(ii) Find the closure of AB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (6+4)     | 2  | L3 |

**SET-2**

**Answer ALL Questions (3\* 10 = 30)**

| Q. No | Questions                                                                                                                                                                                                                                                                 | Max Marks | CO | BL |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|
| 1     | Assume that you are a database administrator at a surgical shop. Is the file processing system suitable to maintain the database of surgical shops? Discuss with justification.                                                                                           | 10        | 1  | L2 |
| 2     | (i) What do you understand about the data anomalies? Explain different types of anomalies might present in the relational database with suitable example.<br>(ii) Reason out why <b>Parent Keys Not Found</b> Error occur and how will you fix it with a suitable example | (6+4)     | 2  | L2 |
| 3     | Let the relation R be R (A, B, C, D, E) and the given set of FDs be F: {A→DC, BC→AD, C→BE, E→A, D→E}.<br>(i) Find the minimal cover of the following set of functional dependencies.<br>(ii) Find the closure of BC.                                                      | (6+4)     | 2  | L3 |

### **SET-3**

**Answer ALL Questions (3\* 10 = 30)**

| Q. No | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Max Marks | CO | BL |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|
| 1     | Describe the three-schema architecture with a neat diagram and distinguish between logical data independence and physical data independence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10        | 1  | L2 |
| 2     | Database to track reviews of papers submitted to an academic conference. Prospective authors submit papers for review and possible acceptance in the published conference proceedings. Consider the following activities and Design a relational model with all possible constraints.<br>(i) An author has valid account information, including their name, id and unique email address.<br>(ii) The paper submitted by the author has information including paper id, authors' name, the title of the article, and review status like pending, accepted or rejected.<br>(iii) A reviewer information, including their id, name, valid email address and specialization.<br>(iv) An author may submit more than one paper in the conference.<br>(v) A reviewer may review one or more papers and submit their report with their decision as accept or reject.<br>(vi) A complete review information must be maintained, including paper id, reviewer id and decision. | (10)      | 2  | L2 |
| 3     | Let the relation R be R (A, B, C, D,) and the given set of FDs be F: {AB→DC, BC→AD, C→D, D→A}.<br>(iii) Find the minimal cover of the following set of functional dependencies.<br>(iv) Find the closure of BC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (6+4)     | 2  | L3 |